Semantic Early-Stopping for Iterative LLM Agent Loops: A Judge-Efficient Study of *When to Halt*

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`semantic-halting-problem`

Working draft — theory claims machine-checked; empirical results from the open harness.

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Abstract

Iterative multi-agent LLM loops—e.g. a Writer that drafts and a Critic that revises—are typically terminated by a fixed iteration cap (`max_iterations`). This *syntactic* kill-switch is blind to whether the answer is still improving: it over-spends on easy inputs and truncates hard ones. We study **semantic early-stopping**: halting when consecutive draft embeddings stop changing in meaning (cosine-distance patience) and the answer's measured quality plateaus. We contribute (i) an *honest* theoretical footing—a proof of deterministic termination and well-definedness, with the convergence of the distance sequence treated as an empirically tested *conjecture* rather than a (previously over-claimed) Banach contraction; (ii) a **judge-efficient evaluation protocol**—generate each trajectory once, replay all stopping policies over it, and cache every LLM-judge call—enabling a strictly paired efficiency-versus-quality comparison on modest compute; and (iii) an empirical study on multi-hop RAG question answering (HotpotQA). On the development split a *judge-free* semantic stopper reduces operational tokens by 37% relative to `max_iterations` at parity quality, while the full quality-gated variant is *counter-productive* because its per-round judging dominates cost. An oracle that selects the best round attains +0.13 Information Score over every practical policy ($p \approx 3 \times 10^{-5}$), reframing the open problem from *when to stop* (easy) to *which round is best* (unsolved).

1 Introduction

A widely used agentic pattern is the **Writer→Critic loop**: a Writer produces an answer, a Critic critiques it, the Writer revises, and the cycle repeats. A termination rule is required. The default—stop after N rounds—is wasteful in two directions. *Over-iteration:* easy questions converge in one or two rounds yet the loop spends tokens and latency until N. *Under-iteration:* hard questions may need more than N and are cut off. Crucially, a counter cannot perceive that rounds 4, 5, and 6 all said essentially the same thing: the decision is made on iteration index, not content.

We map each draft to an embedding and measure the cosine distance between consecutive drafts. When that distance stays below a threshold ε for k consecutive rounds, the answer has converged in meaning, and we halt. We pair this geometric signal with a quality signal (an Information Score over RAG metrics), so the loop halts only when the answer has both stopped changing and stopped improving, with critic approval and a hard failsafe as additional conditions.

Contributions.

1. **Honest theory** (§4): deterministic termination, well-definedness, and halt-priority consistency are *proven and machine-checked*; semantic non-expansiveness is a *measured conjecture*.
2. **Judge-efficient evaluation protocol** (§5): trajectory replay + cached judging + an explicit *operational-versus-evaluation* token model.

3. **Empirical study** (§6) on HotpotQA against five baselines with paired significance and non-inferiority testing.

2 Related Work and Novelty

Semantic convergence and termination of agent loops have been studied from fixed-point theory, multi-LLM uncertainty, adaptive orchestration, contraction mappings in feature manifolds, and token-efficient scheduling. SHP differs in *what* it decides (when to exit a fixed sequential loop, by content) and in its honesty (it proves termination but refuses the unsupported contraction guarantee).

Is this unique? The *ingredients*—early stopping, early-exit inference, embedding similarity—are individually established. What is, to our knowledge, not done together is: (1) *quality-gated* semantic halting for agent loops; (2) a judge-efficient, strictly-paired *evaluation protocol* that makes "which halt policy is best" cheaply measurable; and (3) an *operational-versus-evaluation* token accounting that charges a policy for its own measurement overhead. Contributions (2)–(3) are reusable beyond SHP and constitute the defensible novelty: a methodology, not a new mathematical object.

3 Method

3.1 Architecture

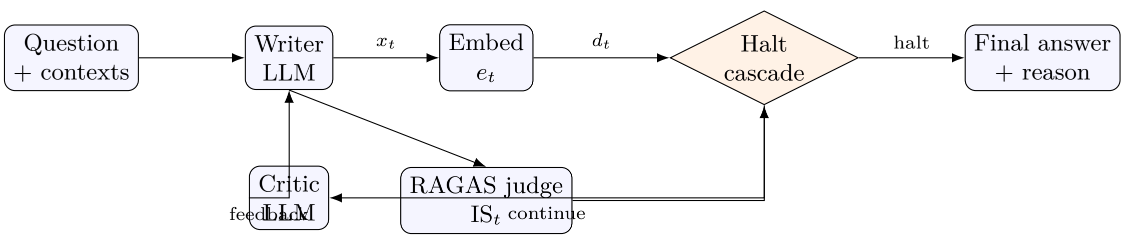


Figure 1: SHP architecture. Embeddings are local (free); only the judge costs API tokens.

3.2 Signals

For drafts $x_1, \ldots, x_T$ with embeddings $e_t = \phi(x_t)$,

$$d_t = 1 - \cos(e_t, e_{t-1}) \in [0, 2], \qquad \mathrm{IS}_t = \sum_{m \in \mathcal{M}} w_m \, \mathrm{RAGAS}_m(x_t) \in [0, 1],$$

with $\mathcal{M} = \{$faithfulness, answer relevancy, context precision, context recall$\}$ and weights w on the simplex.

3.3 The halt cascade

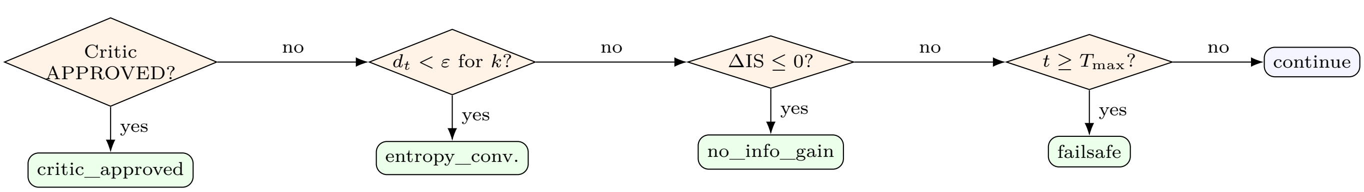


Figure 2: Priority-ordered halt cascade. One shared implementation drives the live loop, the post-hoc reason, and offline replay (Lemma 2).

4 Theory (Honest)

We do *not* claim a Banach contraction: LLM generation has no proven Lipschitz constant < 1 and is non-deterministic across calls. We prove only what holds; each proven claim is machine-checked in the released code.

Theorem 1 (Termination)**.** *For any input, weights, and signal configuration, the loop halts in at most* $T_{\max}$ *rounds.*

Proof sketch. The Critic increments the round counter by one per round; the failsafe clause is unconditional and not ablatable, returning `failsafe` whenever $t \geq T_{\max}$. Hence the stopping time $\tau \leq T_{\max}$. $\square$

Lemma 1 (Well-definedness)**.** $\mathrm{IS}_t \in [0, 1]$ *(convex combination of* $[0, 1]$ *metrics on the simplex);* d_t *is total and bounded in* $[0, 2]$*, with the zero-norm case mapped conservatively to* 1.0*.*

Lemma 2 (Halt-priority consistency)**.** *The post-hoc halt reason equals the live decision, because both invoke a single shared cascade function.*

Conjecture 1 (Semantic non-expansiveness; empirical)**.** *The distance sequence* d_t *tends to decrease in* t*. We* measure *the fraction of monotone trajectories, the mean regression slope (95% CI), and a one-sided Wilcoxon test—reporting null results where they occur.*

5 Experimental Setup

Benchmark. HotpotQA (distractor), filtered to multi-hop `hard` questions so the loop has room to iterate; $N \approx 80$, split 20 dev / 60 test; ε, k tuned on dev only, test frozen; ≈ 4 contexts/question (gold + distractors).

Models. Agents: `llama-3.1-8b-instruct`; judge: 8B on dev, 70B on the frozen test split. Served via an OpenAI-compatible endpoint; embeddings local. Compute is credit-based and modest—a stated limitation.

Protocol. Each question's full trajectory is generated *once*; every halt policy replays over the identical drafts (strictly paired); the judge runs at most once per distinct draft. **Operational** tokens (charged to a policy) include the judge *only if the policy needs it to run*; **evaluation** tokens are a measurement instrument and are not charged.

Policies. `shp` (full cascade), `entropy_only` (judge-free), `critic_only`, `fixed_k`$\in \{1, 3, 6\}$ (`fixed_k6` = `max_iterations`), `random_stop`, `oracle_is` (quality upper bound).

Statistics. Paired t-test, Wilcoxon, Cohen's d_z, bootstrap CIs, TOST non-inferiority on IS (margin δ), Holm correction.

6 Results

6.1 Development split (real; $N = 20$; 8B judge)

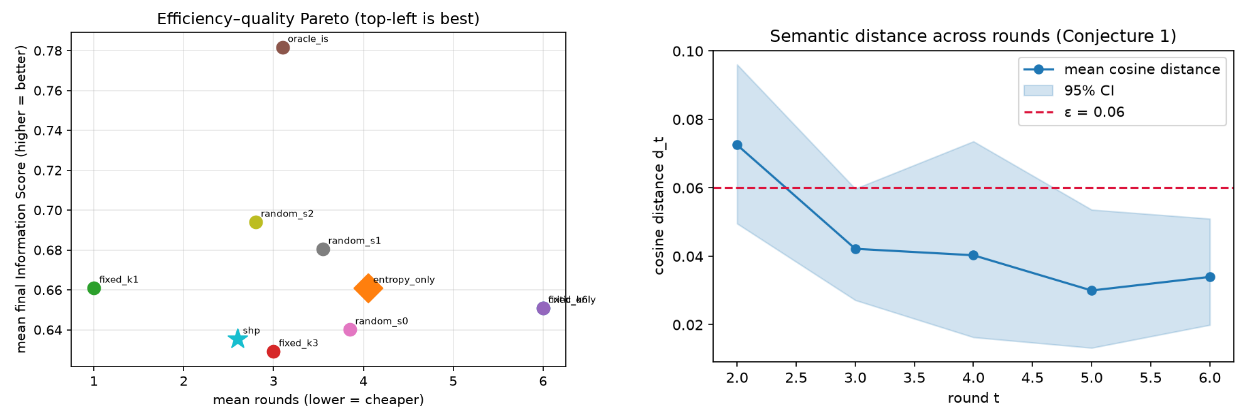
Table 1 reports the real development-split outcome. The baseline is `fixed_k6` (the `max_iterations` policy).

6.2 Test split (frozen; $N = 60$; 70B judge)

Pending—run in progress. Pre-registered prediction: `entropy_only` sits at the top-left of the Pareto front (few rounds, near-baseline quality, no judge cost); full `shp` matches its quality and rounds but pays judge overhead. The frozen-split numbers replace this paragraph.

Policy	Rounds	Op. tokens	Tokens vs base	Final IS
`fixed_k6` (baseline)	6.0	11,281	—	0.651
`entropy_only`	4.05	**7,068**	−37%	**0.661**
`critic_only`	6.0	11,281	0%	0.651
`fixed_k3`	3.0	5,217	−54%	0.629
`fixed_k1`	1.0	**1,548**	−86%	0.661
`shp` (full)	2.6	**27,130**	+140%	0.636
`oracle_is` (ceiling)	3.1	33,007	+193%	**0.782**

Table 1: Development-split results. `entropy_only` saves 37% tokens at parity quality; full `shp` is the most expensive policy for no quality gain; the oracle's +0.13 IS over all practical policies ($p{\approx}3e{-}5$) shows large unrealised headroom. $N = 20$ with an 8B judge is underpowered: non-inferiority is not yet formally confirmed despite positive point estimates.



(a) Efficiency–quality Pareto (dev).

(b) Mean cosine distance vs round.

Figure 3: Left: quality is weakly tied to rounds; the oracle sits far above the practical policies. Right: distances fall quickly (median 0.026) with a heavy tail, so the $k{=}2$ patience window does real work.

6.3 Findings

1. **Efficiency holds.** Judge-free `entropy_only` cuts 37% of tokens versus `max_iterations` while slightly improving mean quality.
2. **Full SHP is counter-productive.** Running the judge every round to power the information-gain signal makes it the most expensive policy (2.4× baseline) for no quality benefit—validating the judge-free design.
3. **Iteration barely moves measured quality here**: `fixed_k1` ties the best policies, yet the oracle reaches 0.782. A better round exists per question but no current signal locates it.

7 Analysis

The data separate a *solved* efficiency problem from an *unsolved* quality problem. Stopping early is cheap and safe (judge-free entropy, guaranteed termination). But the oracle gap (+0.13 IS, $p{\approx}3e{-}5$) shows that cosine distance and IS-gain do not reliably locate the best draft: "when to stop *for efficiency*" is easy; "which round is *best*" is the real open question this study surfaces. A benchmark caveat: HotpotQA answers are short and often answerable from one grounded draft, under-exercising iterative *quality* gains; a long-form task is the right next test.

8 Limitations

Modest N on credit-based compute; an LLM judge is a noisy proxy (mitigated by a stronger judge on the frozen split and non-inferiority rather than equality testing); δ is a modelling choice with reported sensitivity; Conjecture 1 may fail per-trajectory, but Theorem 1 guarantees safety regardless.

9 Conclusion

SHP reframes "when to stop an agent loop" from a blind counter to a content-aware, quality-gated decision backed by an honest termination guarantee and a reusable, judge-efficient evaluation protocol. Its judge-free variant saves tokens at parity quality; its oracle gap names the valuable open problem of best-round identification.

Reproducibility. Seeds, git SHA, and resolved config are stored per run; proven claims are checked by `python -m shp.theory_checks`.

References

[1] Z. Yang et al. HotpotQA: A Dataset for Diverse, Explainable Multi-hop QA. *EMNLP*, 2018.

[2] S. Es et al. RAGAS: Automated Evaluation of Retrieval Augmented Generation. 2023.

[3] Reference set (fixed-point convergence; multi-LLM entropy; adaptive orchestration; contraction attractors; phase-scheduled efficiency)—arXiv IDs to verify before camera-ready.